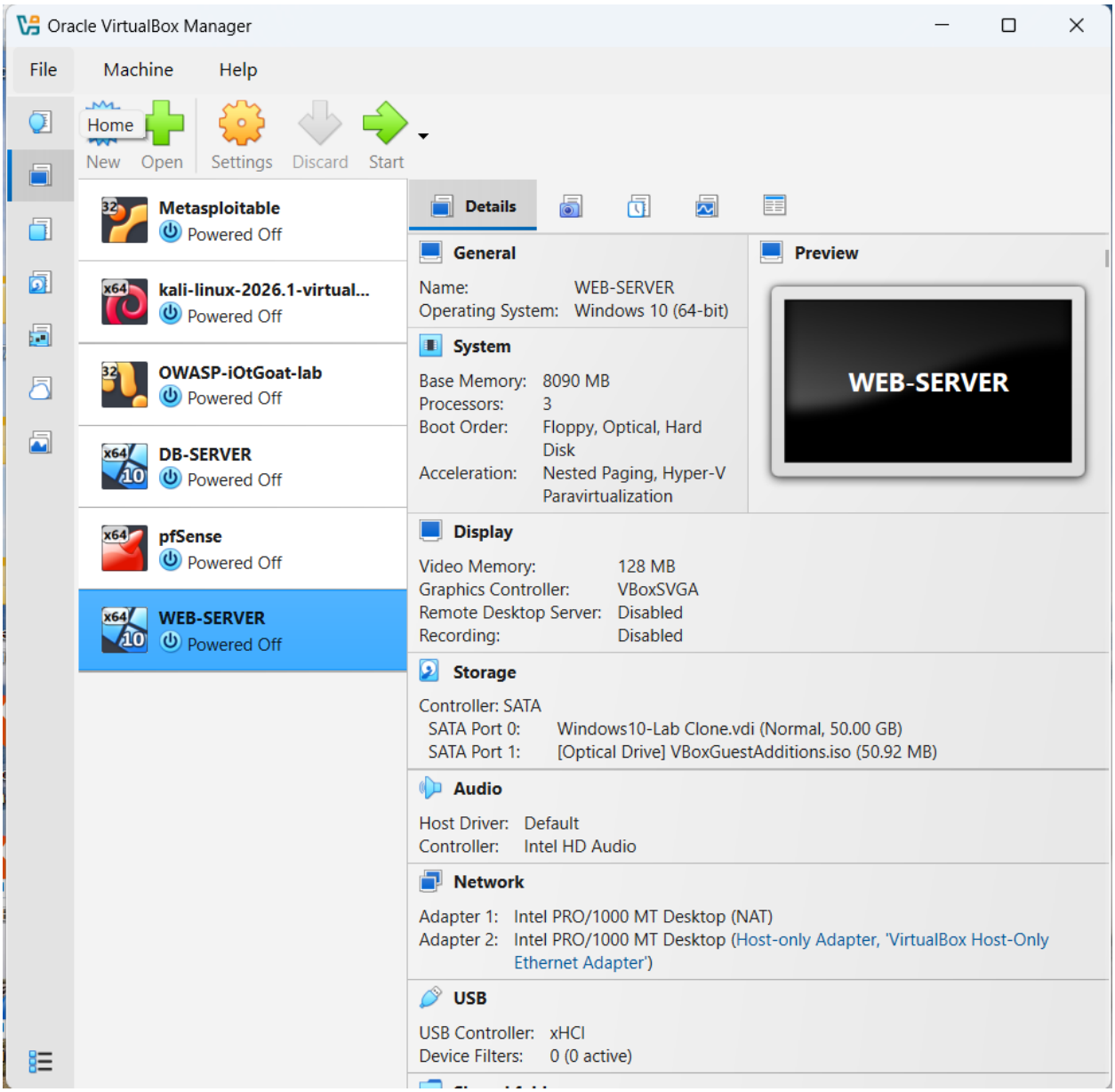
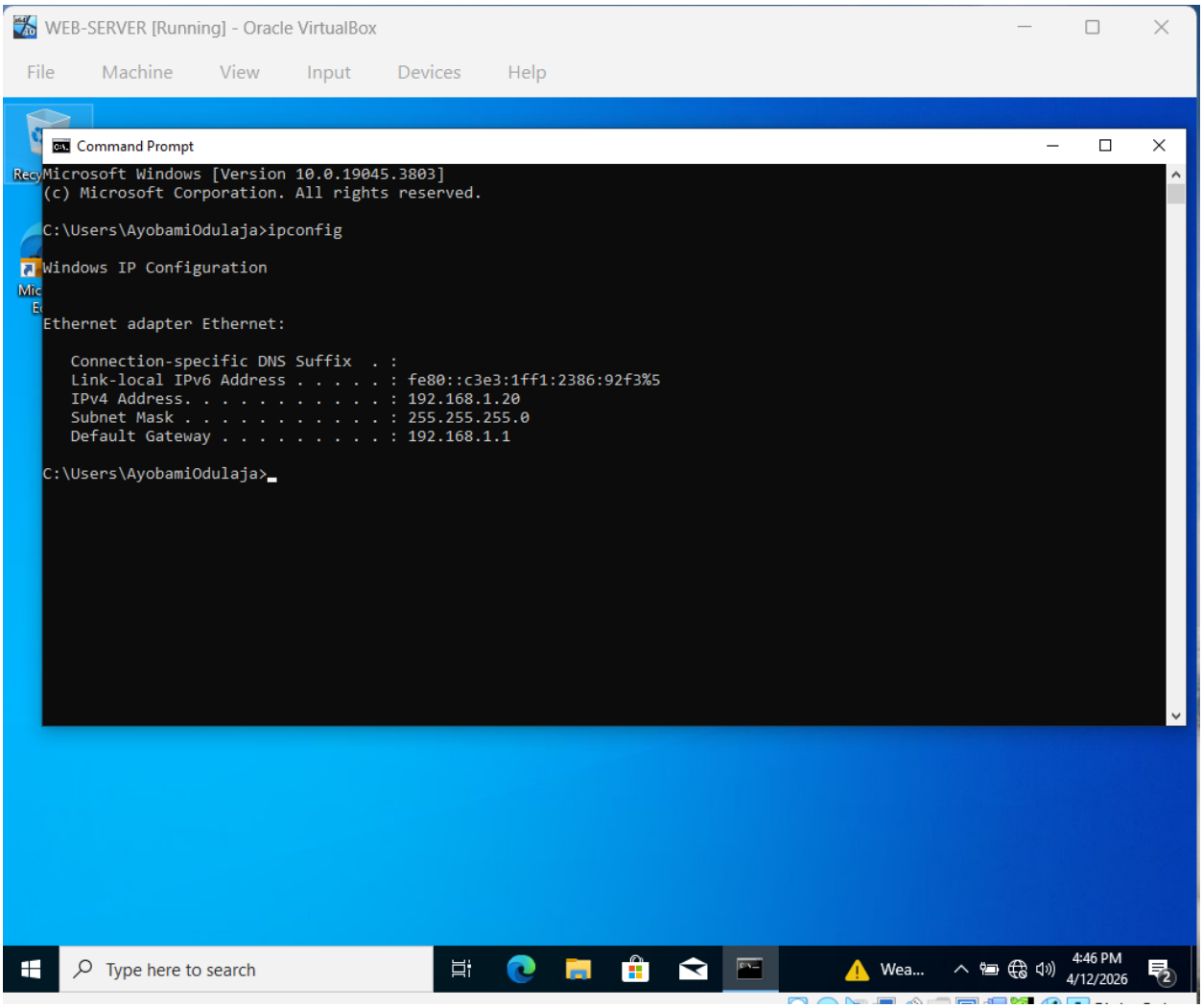
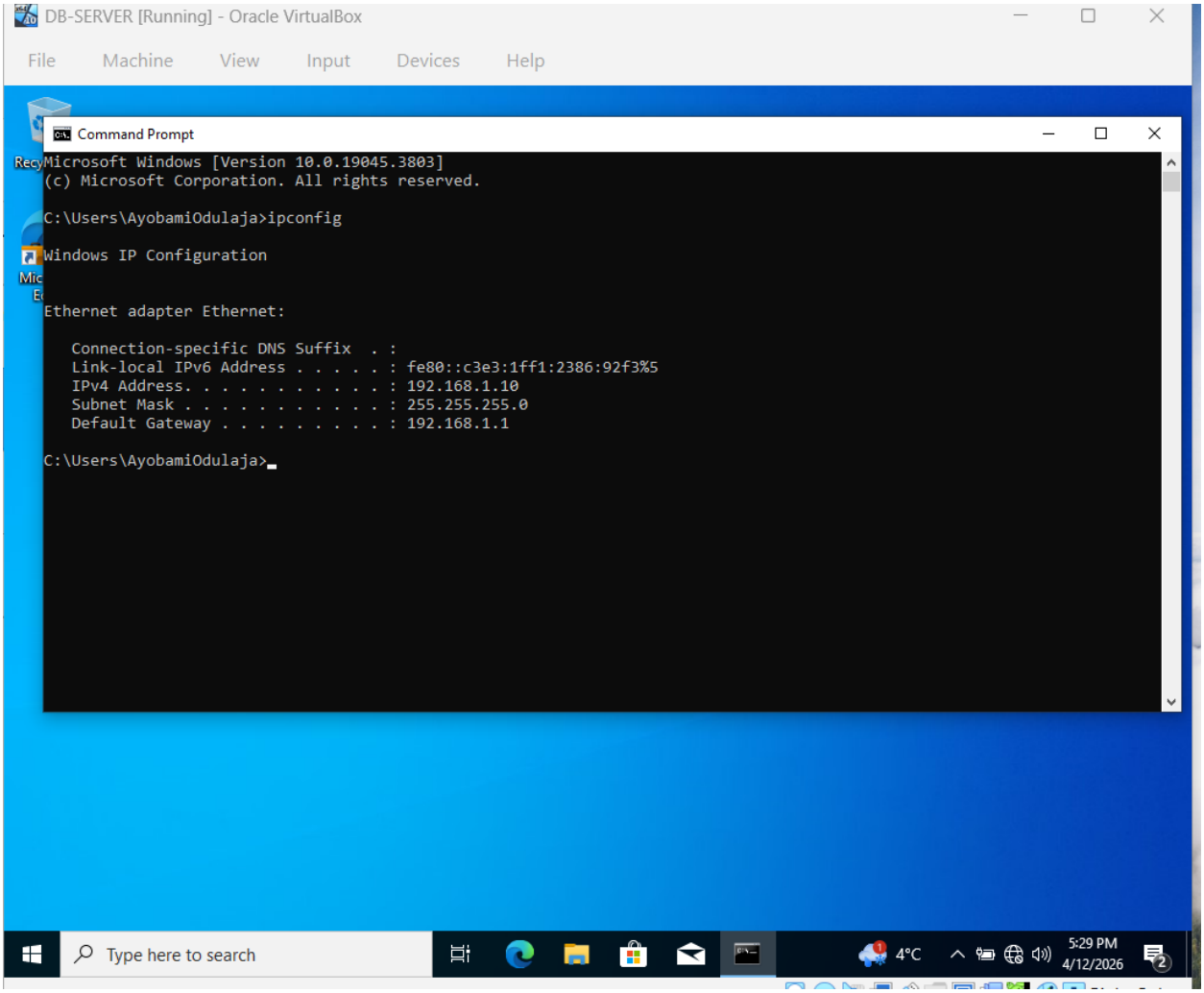
SERVER FIREWALL CONFIGURATION:

VM Set-up

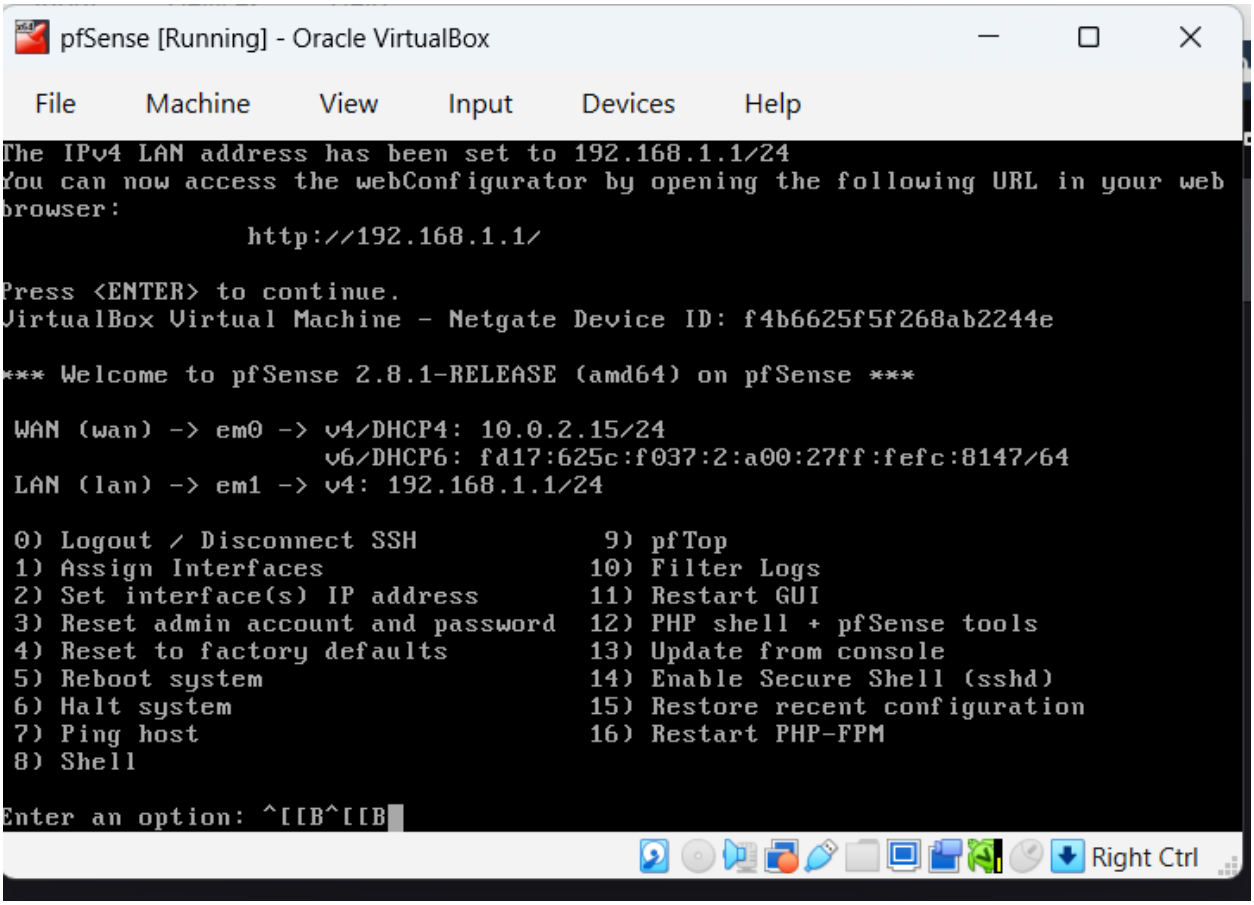


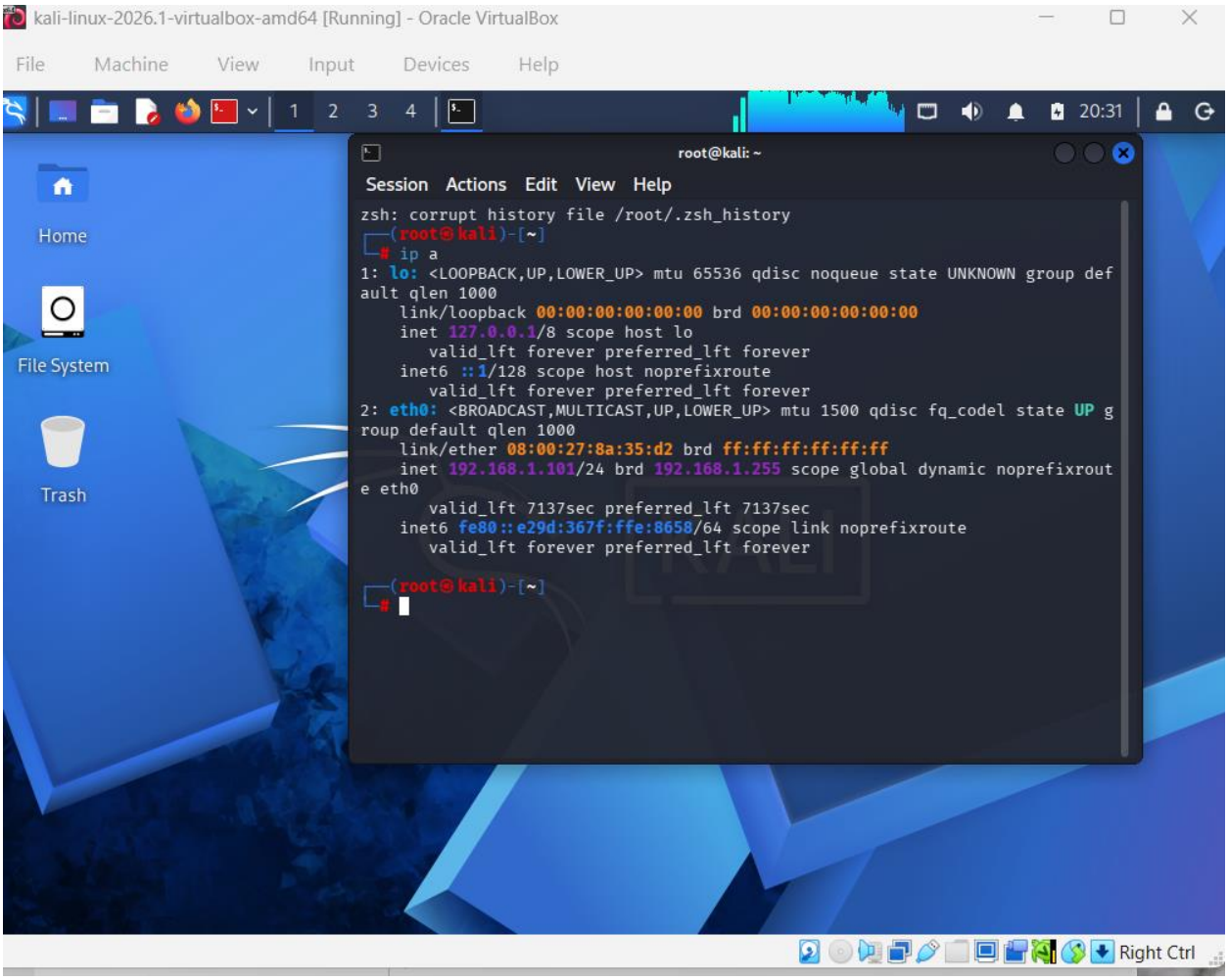
Assigning IP to all VMs:


WEB-SERVER [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:\Users\AyobamiOdulaja>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::c3e3:1ff1:2386:92f3%5
IPv4 Address. . . . . : 192.168.1.20
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
C:\Users\AyobamiOdulaja>
Type here to search
Wea...
4:46 PM
4/12/2026

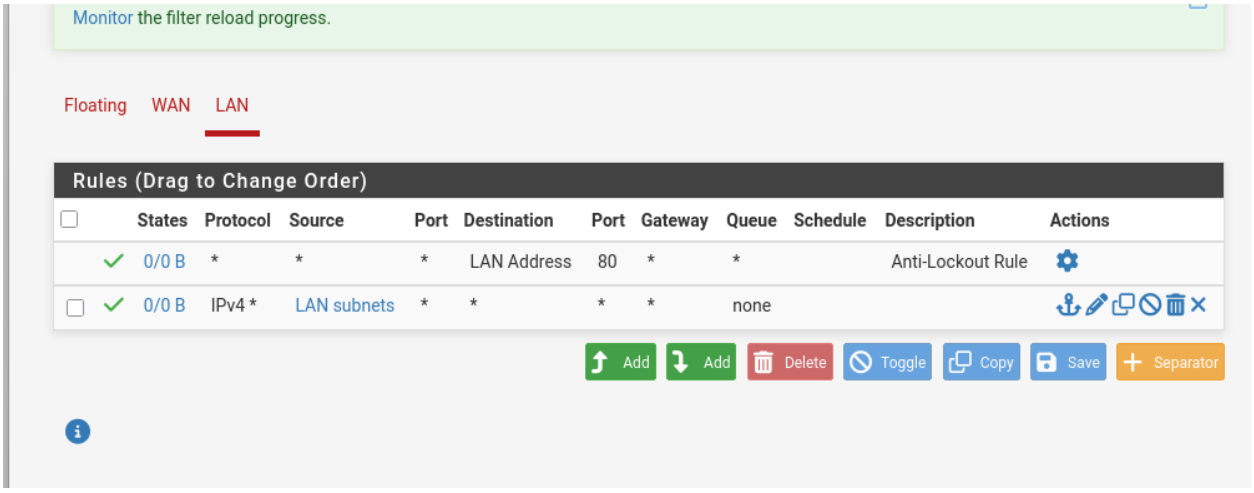

DB-SERVER [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:\Users\AyobamiOdulaja>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::c3e3:1ff1:2386:92f3%5
IPv4 Address. . . . . : 192.168.1.10
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
C:\Users\AyobamiOdulaja>
Type here to search
4°C
5:29 PM
4/12/2026

pfSense IP:


pfSense [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
The IPv4 LAN address has been set to 192.168.1.1/24
You can now access the webConfigurator by opening the following URL in your web browser:
http://192.168.1.1/
Press <ENTER> to continue.
VirtualBox Virtual Machine - Netgate Device ID: f4b6625f5f268ab2244e
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***
WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
v6/DHCP6: fd17:625c:f037:2:a00:27ff:fefc:8147/64
LAN (lan) -> em1 -> v4: 192.168.1.1/24
0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell
9) pfTop
10) Filter Logs
11) Restart GUI
12) PHP shell + pfSense tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM
Enter an option: ^[[B^[[B
Right Ctrl



pfSense Firewall Rule:



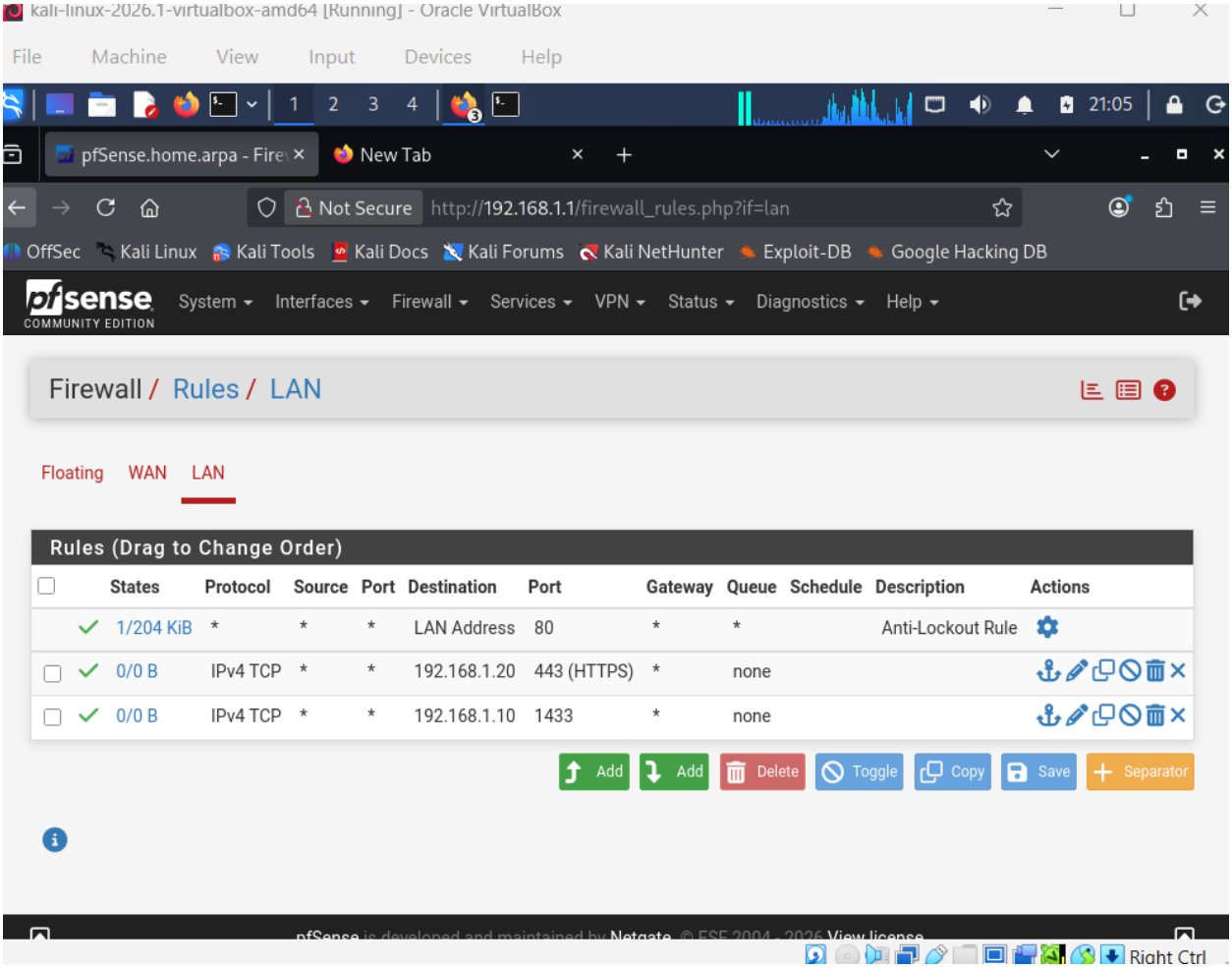
Ping DB-SERVER:

```
root@kali: ~
Session  Actions  Edit  View  Help
--- 192.168.1.20 ping statistics ---
109 packets transmitted, 0 received, 100% packet loss, time 113870ms

┌──(root㉿kali)-[~]
└─# ping 192.168.1.20
PING 192.168.1.20 (192.168.1.20) 56(84) bytes of data.
^C
--- 192.168.1.20 ping statistics ---
12 packets transmitted, 0 received, 100% packet loss, time 11250ms


┌──(root㉿kali)-[~]
└─# ping 192.168.1.10
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.
64 bytes from 192.168.1.10: icmp_seq=1 ttl=128 time=1.81 ms
64 bytes from 192.168.1.10: icmp_seq=2 ttl=128 time=1.02 ms
64 bytes from 192.168.1.10: icmp_seq=3 ttl=128 time=1.02 ms
64 bytes from 192.168.1.10: icmp_seq=4 ttl=128 time=1.04 ms
64 bytes from 192.168.1.10: icmp_seq=5 ttl=128 time=0.758 ms
64 bytes from 192.168.1.10: icmp_seq=6 ttl=128 time=1.06 ms
^C
--- 192.168.1.10 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5005ms
rtt min/avg/max/mdev = 0.758/1.117/1.812/0.326 ms

┌──(root㉿kali)-[~]
└─#
```

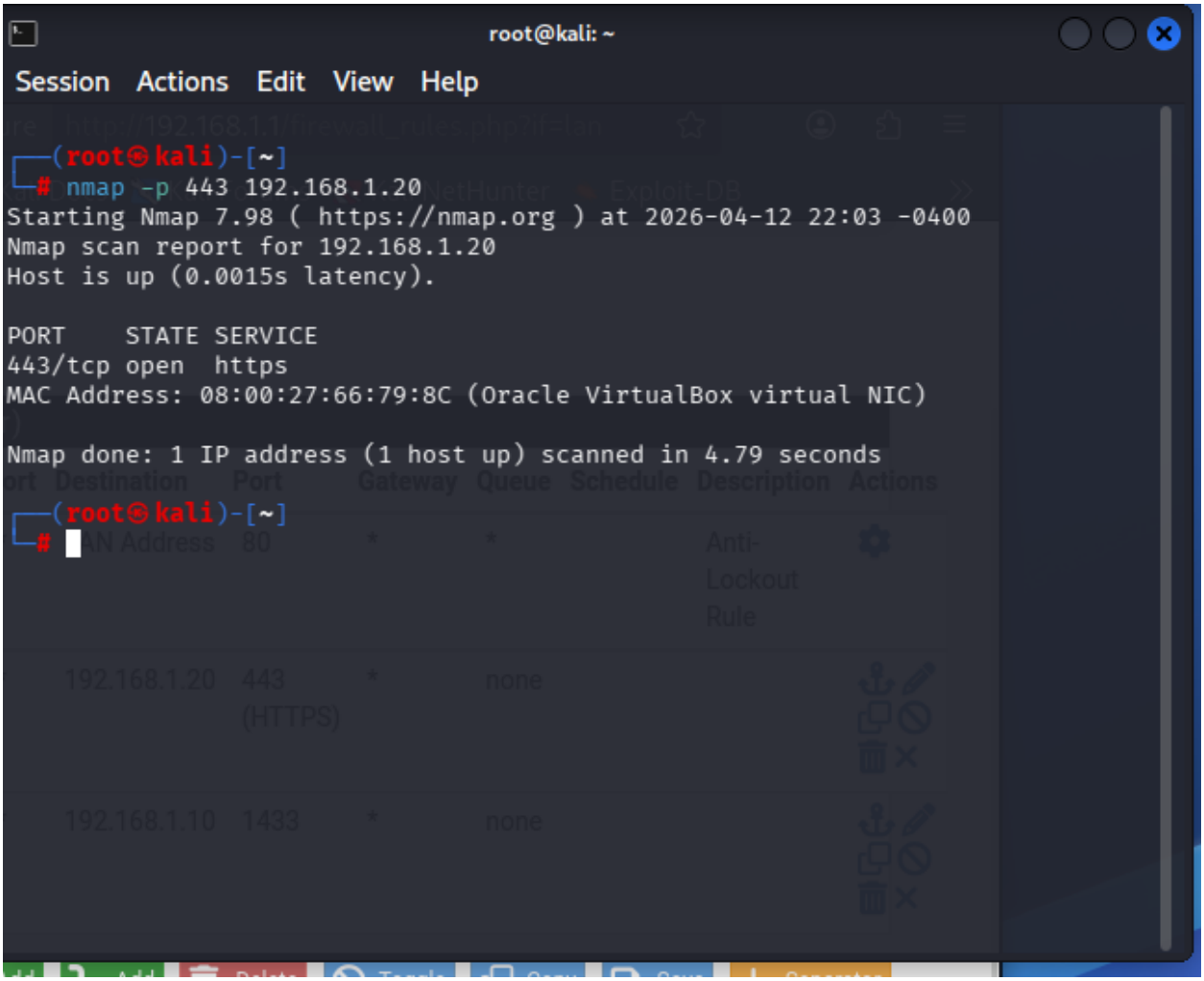
Ping DB-SERVER:

```
root@kali: ~
Session  Actions  Edit  View  Help
└─# ping 192.168.1.10
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.
64 bytes from 192.168.1.10: icmp_seq=1 ttl=128 time=1.81 ms
64 bytes from 192.168.1.10: icmp_seq=2 ttl=128 time=1.02 ms
64 bytes from 192.168.1.10: icmp_seq=3 ttl=128 time=1.02 ms
64 bytes from 192.168.1.10: icmp_seq=4 ttl=128 time=1.04 ms
64 bytes from 192.168.1.10: icmp_seq=5 ttl=128 time=0.758 ms
64 bytes from 192.168.1.10: icmp_seq=6 ttl=128 time=1.06 ms
^C
--- 192.168.1.10 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5005ms
rtt min/avg/max/mdev = 0.758/1.117/1.812/0.326 ms

┌──(root㉿kali)-[~]
└─# ping 192.168.1.20
PING 192.168.1.20 (192.168.1.20) 56(84) bytes of data.
64 bytes from 192.168.1.20: icmp_seq=1 ttl=128 time=3.21 ms
64 bytes from 192.168.1.20: icmp_seq=2 ttl=128 time=0.674 ms
64 bytes from 192.168.1.20: icmp_seq=3 ttl=128 time=0.965 ms
64 bytes from 192.168.1.20: icmp_seq=4 ttl=128 time=4.14 ms
64 bytes from 192.168.1.20: icmp_seq=5 ttl=128 time=1.87 ms
^C
--- 192.168.1.20 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4018ms
rtt min/avg/max/mdev = 0.674/2.171/4.142/1.322 ms

┌──(root㉿kali)-[~]
└─#
```

Create Firewall Rules (pfSense) : For WEB and DB-SERVER



Scan DB-Server:

```
root@kali: ~
Session  Actions  Edit  View  Help
1433/tcp closed ms-sql-s

Nmap done: 1 IP address (1 host up) scanned in 4.74 seconds

┌──(root㉿kali)-[~]
└─# nmap -sT -Pn -p 1433 192.168.1.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-12 21:48 -0400
Nmap scan report for 192.168.1.10
Host is up (0.0088s latency).

PORT     STATE  SERVICE
1433/tcp closed ms-sql-s

Nmap done: 1 IP address (1 host up) scanned in 4.93 seconds
┌──(root㉿kali)-[~]
└─# nmap -sT -Pn -p 1433 192.168.1.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-12 21:57 -0400
Nmap scan report for 192.168.1.10
Host is up (0.00069s latency).

PORT     STATE SERVICE
1433/tcp open  ms-sql-s

Nmap done: 1 IP address (1 host up) scanned in 4.62 seconds

┌──(root㉿kali)-[~]
└─#
```

Scan WEB-SERVER: